

CHAPTER 18

# ENERGY: FORMS AND SOURCES

## 18.1 What is energy?

We use energy to carry out different tasks. In order to complete a task, we need to do some work. We need to work harder to lift a heavy object and it requires more energy.

We have already talked about work and energy in class 9th. We saw that there is a direct relation between work and energy. More energy is required in order to perform more work. The more energy you have, the more work you will be able to do.

The term 'Energy' was introduced 200 years ago by a British scientist, Thomas Young. Young derived the term 'energy' from the Greek word 'evegyeia energia' which means internal energy. According to Young, work is inherent in energy. We can use this energy to perform some work. Even our bodies have energy but in limited quantity. The more work we do, the more energy will be consumed and we will feel tired. We need food to supply us energy.



**Thomas Young**



**Figure-1**

**Human Beings and Energy**

We tend to associate the term 'work' with living human beings. When human beings were cave dwellers, they used to primarily use the chemical energy present in their bodies. Then people

acquired new knowledge. They started harnessing energy from their domestic animals. They learnt how to use the flow of water in rivers to sail boats. They also started using fire as a source of energy by burning woods and animal fat. First, people used fire to get warm during winters and as a source of light at night-time. Then, people started using fire to cook food and to prepare pots of metals, glass or clay. So now they were using the energy, not just from living sources like human-beings or animals but also from non-living sources.

Steam engine was invented in 1700. Different kinds of steam engines were developed and for the first time chemical energy of fuel could be converted into kinetic energy. Some steam engines were used to run machines in industries, some were used for sailing ships across seas and oceans, and some were used to pull locomotives on land.

Coal replaced wood as a fuel. In 1800, chemical energy obtained from combustible fuels was used to turn wire-wheels between the poles of two magnets. In this experiment, the dynamic energy of magnets was used to produce an electric current. Electric current was used in telegraphs and telephones. All the motors in the world run on electricity.

In around 1800 people started digging oil-wells. Oil could be easily transported from one place to another via pipes. At the end of the nineteenth century, petrol-engines were invented. There were used in cars, trucks, buses, ships, aeroplanes etc. Petrol is now also used to produce electricity and in heating homes in colder climates.

There are some other sources of energy apart from oil and coal. We have nuclear energy; energy obtained from the flow of water and wind; energy due to tidal waves (high/low tides) of sea are possible sources of energy. The energy hidden in the earth's core can also be used. Many needs can be fulfilled by solar energy.

In this chapter we will talk about these sources of energy and learn about different types of energy. We will also see what kind of energy can be obtained from a particular source of energy.

## 18.2 Types and forms of energy

Every day we come across various forms of energy. For example, you have breakfast (chemical energy) in the morning; go to school on a bicycle (mechanical energy); a room full of light (light energy); talk to your friends and listen to them (sound energy); throw up a ball up and it comes back into your hands (gravitational energy); enjoy a fan's air (electric energy) etc.

Can you think of any other form of energy? All forms of energy can be classified into two types - kinetic energy and potential energy.

## Forms of energy

### Potential Energy

Stored energy and energy by virtue of position. (Gravitational)

### Kinetic Energy

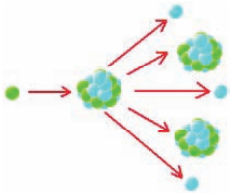
There is energy in motion. This energy is due to motion of waves, electrons, atoms and molecules.

### Chemical Energy

Chemical energy is the energy stored in the bonds of chemical compounds (atoms and molecules). Bio-mass, petroleum, natural gas, propane, coal are some examples of reservoirs of chemical energy.



### Nuclear Energy



The energy stored in the nucleus of atoms is called nuclear energy. It holds together the atom. The nucleus of uranium atoms is an example of nuclear energy.

### Stored Kinetic Energy



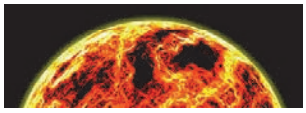
The energy stored in objects due to applied force is called stored kinetic energy. Compressed spring and elongated rubber are examples of stored kinetic energy.

### Gravitational Energy



The energy of an object due to its position is gravitational energy. For example, the energy of water stored in the reservoir of a hydro-electric dam. When this water is released, it moves turbines and is converted to kinetic energy.

### Radiation Energy



Radiation energy is a type of electro-magnetic energy which creates motion in transverse waves. Visible light, x-ray, gamma rays and radio waves as well as solar energy are examples of radiation energy.

### Thermal Energy



Thermal energy is the internal energy of matter. This energy is generated due to the vibrations and motion of atoms and molecules in matter.

### Motion Energy



Movement of an object or material from one place to another is motion. Wind and hydro-electric are all examples of motion.

### Sound Energy



Sound is the movement of energy through substances in longitudinal (compression/rarefaction) waves.

### Electric Energy



Lightening during storms and electricity are examples of electric energy. Electricity is the flow of moving electrons.

## 18.3 Energy Transformations

Motionless wind, stationary water or a stationary stone do not have the capability to move or break any object. We have seen that a non-living object is certainly in motion whenever it performs some work. A strong wind-storm can cause a tree to fall, moving water can push forward a sail-boat, and a heavy stone thrown using a catapult can damage a wall in its path.

Since moving objects result in some work therefore there must be some energy in motion. This energy is called kinetic energy. We have already learnt a little about kinetic energy in previous classes.

You may know that if two objects are moving at the same speed then the heavier amongst the two will have more energy as compared to the lighter object. If we lightly hit a nail with a hammer, it will not go as deep into the wall as compared to the hole formed when a nail is hit hard by a heavy hammer. But is it possible for an object at rest (not in motion) to perform some work?

Let us think about a stone on the top of a mountain-peak. A heavy wind can put it in motion. As it rolls down towards the ground it gains kinetic energy. The speed of a falling body keeps increasing with time. When the stone finally hits the ground, it performs some work; for example, it can form a big hole or it can shatter an object. More the height from ground of an object, the more distance it will have to fall and the more will be its potential energy.

Here, we need to understand that kinetic and potential energy are of many types and all the forms of energy (about which we learnt in previous classes or other chapters) can be classified into these two types.

Every object has some energy stored in it, but we cannot say anything about it till something happens to transform the stored energy into some other form. For example, when we fire a cracker, the chemical energy stored in it gets converted into light (energy), sound (energy), thermal energy and kinetic energy.

Try to fill the following table based on the conversion/transformation of energy from one form to another-

**Table-1**

| **Object** | **What is needed in order to convert the stored energy to some other form** | **Initial form of energy (stored in object)** | **The energy (stored in object) gets converted to the following forms** |
|---|---|---|---|
| 1. Torch Battery | Turning on switch | Chemical energy | Light and electric energy |
| 2. Petrol | | | |
| 3. Dynamite | | | |
| 4. Stretched rubber band | | | |
| 5. Tabla | | | |

You can think of more examples based on the initial and final forms of energy. Try to fill the following table-

**Table-2**

| **ENERGY** | | **Example** |
|---|---|---|
| **Initial form** | **Final form** | |
| Potential energy | Kinetic energy | • Water falling from tank from a higher level.<br>•<br>• |
| Kinetic energy | Thermal energy | • Palms get warm on rubbing.<br>•<br>• |
| Kinetic energy | Potential energy | • Compressing a spring.<br>•<br>• |
| Kinetic energy | Electric energy | • Production of electricity using electric generator<br>•<br>• |
| Chemical energy | Light energy | • Switching on torch.<br>•<br>• |
| Kinetic energy | Sound energy, light energy, thermal energy | • Knife hitting a stone.<br>•<br>• |
| Electric energy | Thermal energy | • Iron getting warm.<br>•<br>• |

Do you know that the energy always getting converted from one form to another in this world? Not only this, energy is always getting transferred from one object to another.

All electric appliances which we use, whether battery-operated or running on electricity, convert electric energy to some other form of energy. Most of the energy gets converted into useful energy but some amount gets wasted or converted into a form which is not useful. Irrespective of whether the transformed form of energy is useful or waste, the entire electrical energy gets converted. This is known as 'law of conservation of energy'. In other words energy can neither be created nor destroyed; it can only be transformed from one form to another.

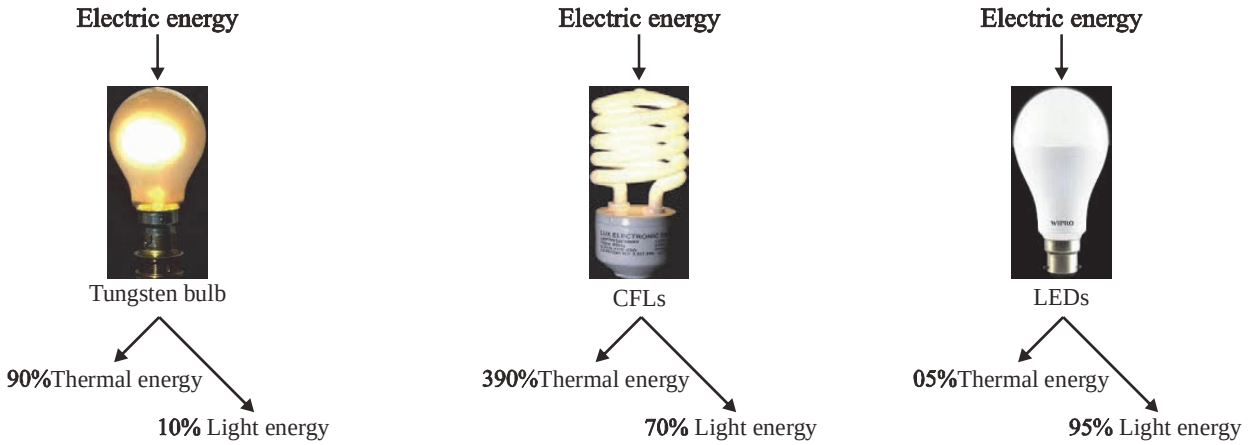
The following table consists of some such examples of energy conversion where the electrical energy gets converted into both useful and useless forms while using electric appliances. Try to add more such examples to the table-

**Table-3**

| S.No. | Electrical appliance | Form into which electrical energy is converted | |
|---|---|---|---|
| | | **Useful** | **Waste** |
| 1. | T.V. | Light and sound energy | Thermal energy in environment and T.V. |
| 2. | Hair dryer | Kinetic and thermal energy of air molecules | Sound energy |
| 3. | Oven | Thermal energy (food gets warm) | Thermal energy (utensils become hot), sound energy |
| 4. | Fan | Kinetic energy | Sound energy |
| 5. | | | |

We cannot eliminate this loss of energy but this waste can be minimized.

You have read about tungsten bulbs in a previous chapter. These bulbs convert electric energy into light energy and thermal energy but the thermal energy is not useful. The loss of energy can be minimized using CFLs (Compact Fluorescent Lights) or LEDs (Light Emitting Diodes). LEDs are even better than CFLs because LEDs convert a higher percentage of electric energy into light energy.




## 18.4 Energy transfer

In the chapter on heat and temperature, we read about transfer of thermal energy. Thermal energy can be transferred in three ways - conduction, convection and radiation. Knowing how thermal energy is transferred from hotter objects to colder objects can help us control loss of thermal energy.

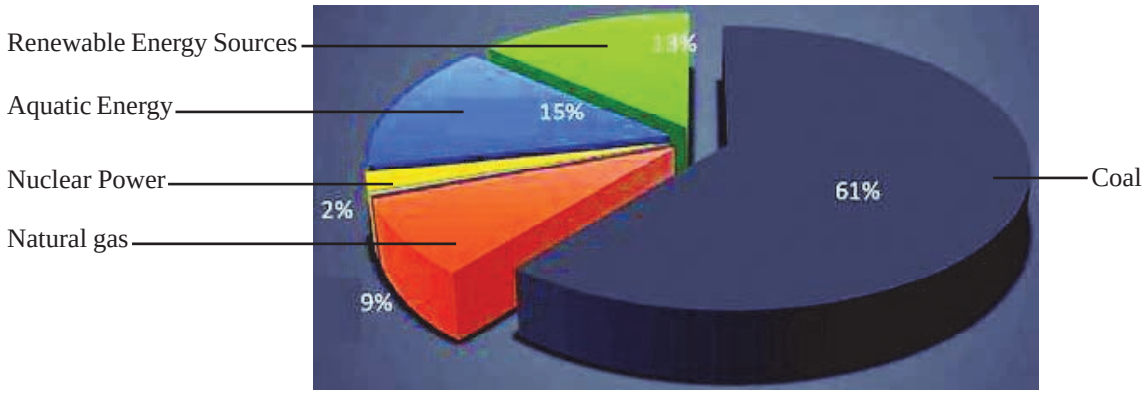
## 18.5 Sources of energy

No matter what form of energy we need, we first have to obtain it from some source. For example, the electrical energy in our homes could be from coal, hydro-electric power plants or nuclear fission. The chemical energy to drive vehicles can be obtained from fuels like petrol, diesel etc. These are called sources of energy.

Today, what our country need most are sources of energy which can fulfill increasing demand of electric energy.

To produce electricity, we need to convert energy from one form to another. The method could be direct conversion (which converts the energy directly to electricity) or an intermediate conversion (converting energy to some form and then converting that form to electricity).

For both of these methods different kinds of energy sources are used like fossils fuels, nuclear fission, solar energy, hydro-electric power etc.




Central Electricity mission, based on the report of the Government of India figures

**Figure-2: Different sources of electricity in India**

### 18.5.1 Conventional sources of energy

**1. Fossils fuels:** Coal, petroleum, LPG, kerosene, diesel etc. are fuels. There are directly used in vehicles and gas stoves. These are also used as a primary source for generating electricity. Large amount of fossil fuels are burnt every day in power stations to heat up water to produce steam which further runs the turbine to generate electricity. These appliances are called thermal appliances because these appliances generate energy by combustion of fuels which is then converted into electric energy.

These appliances convert chemical energy of fuels into thermal energy, which then gets converted into mechanical energy (kinetic energy) by rotating turbines and finally, using electric generators, this is converted into electric energy. At every step, some amount of energy is lost as it gets converted into a useless form. Power plants that run on fossil fuels usually have an efficiency of 40% which means that only forty percent of the energy of fossil fuels gets converted into electrical energy.

There are some other disadvantages of burning fossils fuel. Oxides of carbon, nitrogen and sulphur are released into the atmosphere on burning fossils fuel and they cause acid rain which pollutes water and can harm monuments.

**2. Hydro-electric power plants:** Another conventional source of energy is the kinetic energy of flowing water or the potential energy of water at a height. Hydro power plants convert the potential energy of falling water into electricity. The water falls on a turbine which spins and activates an electric generator which then produces electricity.

In order to produce hydel electricity, high-rise dams are constructed on the river to obstruct the flow of water and thereby collect water in larger reservoirs. During the rainy season, water collects in these reservoirs. The stored water from the high level in the dam is carried through pipes, to the turbine, at the base of the dam. Stored water falls freely on the blades of turbine and as a result, the blade of turbine spin and with the help of a generator, electricity is generated. Hydel-electricity is a renewable source of energy because after rains, the water reservoirs get refilled. There are also several disadvantages associated with hydro-electric power plants. During the construction of dams and water reservoirs, huge amounts of agricultural land is flooded and lost and many people lose their homes and are displaced. Some ecological systems are also destroyed.

**3. Bio gas:** Agriculture and cattle-rearing are given great importance in our country. Cow-dung (gobar) is available in steady supply due to large number of livestock and agricultural waste in our country. Cow-dung cakes are burnt directly and used as fuels in many rural areas but a large part of energy is lost in this process. Also these fuels do not produce much heat on burning and a lot of smoke is given out when they are burnt. Therefore, technological inputs to improve the efficiency of these fuels are necessary.

When wood is burnt in a limited supply of oxygen, water and volatile materials present in it get removed and charcoal is left behind as the residue. Charcoal burns without flames, is comparatively smokeless and has higher heat generation efficiency.

Cow-dung, various plant materials like the residue after harvesting the crops, vegetable waste and sewage are decomposed in the absence of oxygen to give bio-gas. Bio-gas is a mixture of methane, carbon dioxide, hydrogen and hydrogen sulphide.

Bio-gas plant has a dome-like mixing-tank built with bricks in which a slurry of cow-dung and water is made. The slurry is fed into the digester which is a sealed chamber in which there is no oxygen. Anaerobic micro-organisms that do not require oxygen decompose or break down complex compounds of the cow-dung slurry. It takes a few days for the decomposition process to be complete and generate bio-gas. The bio-gas is stored in the gas tank above the digester from which they are drawn through pipes for use. Bio-gas is an excellent fuel as it contains up to 75% methane. It burns without smoke and leaves no residue. Bio-gas is also used for lighting.

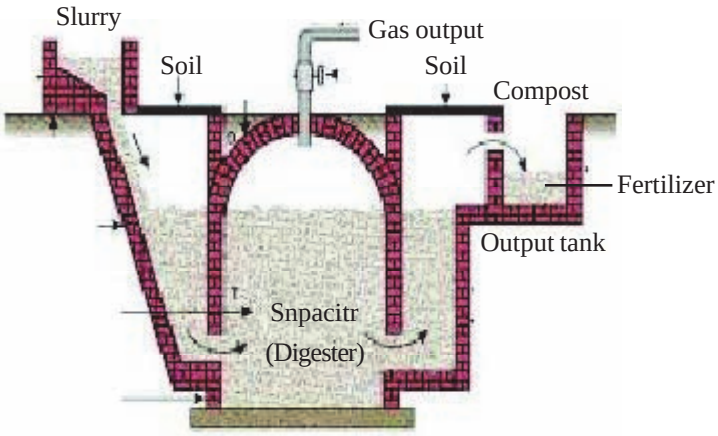
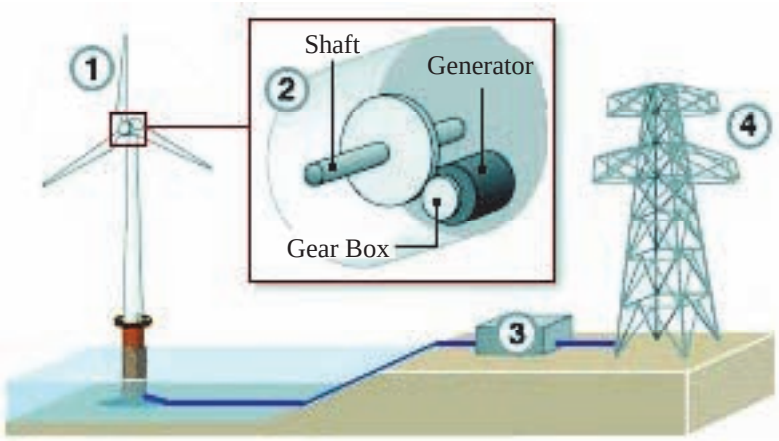



**Figure-3**

**4. Wind Energy:** We saw in a previous chapter how unequal heating of the landmass and water bodies by solar radiation generates air movement and causes winds to blow. This kinetic energy of the wind can be harnessed by windmills to do mechanical work. For example, in a water-lifting pump, the rotatory motion of windmill is utilised to lift water from a well. Today, wind energy is also used to generate electricity.

To generate electricity, the rotatory motion of the windmill is used to turn the turbine of the electric generator. A number of windmills are erected over a large area, which is known as wind energy farm. Wind energy is renewable source of energy. But, wind energy farms can be established only where the speed of wind is higher than 15 km/h to maintain the required speed of the turbine.




1. Wind causes the wind-mill turbine to rotate.
2. The shaft activates the generator which produces electricity.
3. Transformer transforms electricity at high potential.
4. Electricity grid transmits electricity.

**Figure-4**

### 18.5.2 Alternate sources of energy

Our demands for energy are increasing everyday due to progress in technologies. It is necessary to develop new techniques for the efficient uses of energy sources and keep on researching for new sources of energy.

**1. Solar energy:** India is fortunate to receive solar energy for greater part of the year. By solar energy, we mean the light and heat energy which falls on the earth. Sun is a huge source of energy. Solar energy is very different from the conventional sources of energy. Electric energy can be generated by direct conversion of solar energy.

**Solar heating devices:** The devices which can store solar energy are called solar heating devices.

**Procedure:**
- Take two conical flasks and paint one white and the other black. Fill both with water.
- Place the conical flasks in direct sunlight for half an hour to one hour.
- Touch the conical flasks. Which one is hotter? You could also measure the temperature of the water in the two conical flasks with a thermometer.
- Can you think of ways in which this finding could be used in your daily life?

**Solar cooker:** Solar cooker is a device which is used to cook food using the radiation heat energy of the sun. Solar cooker box is made up of wood box or some other insulating material. The inner body of a solar cooker is painted black. The food to be cooked is put into metal pots, whose outsides are painted black. Then the metal pots are put inside the solar cooker box which is covered using a glass sheet.

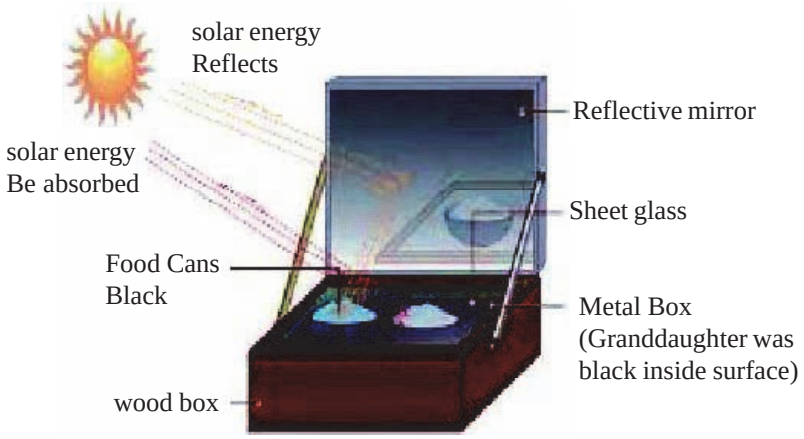



**Figure-5**

Once the heat waves of sun enter the box, the glass cover does not allow them to escape out of box easily. This way, maximum energy of the heat waves is captured inside the box due to which the temperature of solar cooker box increases to 100-140° C in 2-3 hours. This heat is used to cook edible foods like, rice, pulses and vegetables inside the black pots.

**Solar Cells**

It is easy to see that these devices are useful only at certain times during the day. This limitation of using solar energy is overcome by using solar cells that convert solar energy into electricity. A typical cell develops a voltage of 0.5-1 V and can produce about 0.7 W of electricity when exposed to the Sun. A large number of solar cells are, combined in an arrangement called solar cell panel that can deliver enough electricity for practical use.

The principal advantages associated with solar cells are that they have no moving parts, require little maintenance and work quite satisfactorily without the use of any focussing device. Another advantage is that they can be set up in remote and inaccessible areas or very sparsely inhabited areas in which laying of a power transmission line may be expensive and not commercially viable.



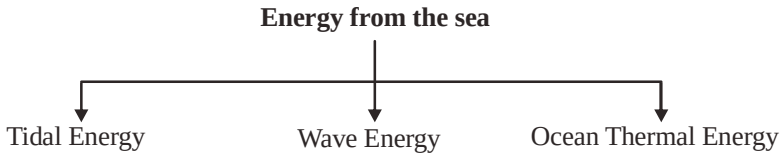
**Figure-6**

Silicon, which is used for making solar cells, is abundant in nature but availability of the special grade silicon for making solar cells is limited. The entire process of manufacture is still very expensive, silver used for interconnection of the cells in the panel further adds to the cost. In spite of the high cost and low efficiency, solar cells are used for many scientific and technological applications.

- Artificial satellites and space probes like Mars orbiters use solar cells as the main source of energy.
- Radio or wireless transmission systems or TV relay stations in remote locations use solar cell panels.
- Traffic signals, calculators and many toys are fitted with solar cells. The solar cell panels are mounted on specially designed inclined roof tops so that more solar energy is incident over it.

The domestic use of solar cells is, however, limited due to its high cost.

**2. Energy from the sea**




(a) Tidal energy: Due to the gravitational pull of mainly the moon on the spinning earth, the level of water in the sea rises and falls. The difference in sea-levels during high and low tides gives us tidal energy. Tidal energy is harnessed by constructing a dam across a narrow opening to the sea. A turbine fixed at the opening of the dam converts tidal energy to electricity.

(b) Wave energy: Similarly, the kinetic energy possessed by huge waves near the seashore can be trapped in a similar manner to generate electricity. The waves are generated by strong winds blowing across the sea.

(c) Ocean thermal energy: The water at the surface of the sea or ocean is heated by the Sun while the water in deeper sections is relatively cold. This difference in temperature is exploited to obtain energy in ocean-thermal-energy conversion plants.

**3. Geothermal energy:** Due to geological changes, molten rocks formed in the deeper hot regions of earth's crust are pushed upward and trapped in certain regions called 'hot spots'. When

underground water comes in contact with the hot spot, steam is generated. Sometimes hot water from that region finds outlets at the surface. Such outlets are known as hot springs. Sometimes the steam gets trapped in rocks and its pressure increases. This steam is routed through a pipe to a turbine, the turbine spins and activates a generator and produces electricity. The cost of production would not be much, but there are very few commercially viable sites where such energy can be exploited. There are number of power plants based on geothermal energy operational in New Zealand and United States of America.

**4. Nuclear energy:** We have already learnt that the nucleus of an atom consists of neutrons and protons, and electrons are orbit around the nucleus. Scientists have discovered that the nucleus inside the atoms are a huge source of energy, therefore this energy is called nuclear energy. Nuclear energy could be obtained (produced) using two following reactions.

(a) Nuclear fission (b) Nuclear fusion

**4 (a) Nuclear fission:** Nuclear fission is a reaction the nucleus of a heavy atom (such as uranium, plutonium or thorium), when bombarded with low-energy neutrons, can be split apart into lighter nuclei. If the mass of the original nucleus is just a little more than the sum of the masses of the individual products after bombardment, a huge quantity of energy is released. The fission of an atom of uranium, for example, produces 10 million times the energy produced by the combustion of an atom of carbon from coal. In a nuclear reactor designed for electric power generation, such nuclear 'fuel' can be part of a self-sustaining fission chain reaction that releases energy at a controlled rate. The released energy can be used to produce steam and further generate electricity.

**Chain Reaction:** When 92U235 is bombarded by slowly moving neutrons, each uranium nucleus breaks into two almost equal parts and gives rise to huge amount of energy and three neutrons. Under favorable conditions, these neutrons can further divide other uranium nuclei. In this way, a fission chain is initiated which once started continues till all the Uranium nuclei have undergone fission. This process of fission of nucleus is called chain reaction.

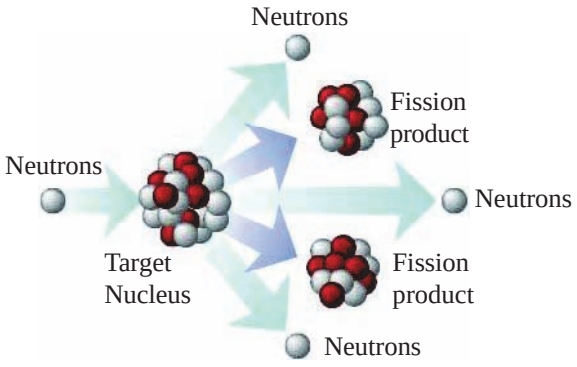



**Figure-7**

Chain reactions are of two types:

1. Uncontrolled chain reaction 2. Controlled chain reaction

**4.a.1 Uncontrolled chain reaction:** When U-235 bombarded by slowly moving neutrons, it results in formation of three new neutrons. Each of these three neutrons can again cause the fission of more U-235 nuclei and produce more neutrons. This fission process proceeds at a very fast rate and the atoms of the entire sample undergo fission in a very short time. A huge amount of energy is released in this reaction and results in a huge explosion. This reaction is used in nuclear bombs.

**4.a.2 Controlled chain reaction:** Nuclear energy is also used for constructive purposes. To obtain controlled amount of energy, artificial means are used to ensure that only one out of the three neutrons produced can further carry out the fission reaction. In this way, the rate of nuclear fission remains constant and the reaction is slowed down. Nuclear reactors make use of controlled fission reactions for energy production.

One major hazard of nuclear power generation is the storage and disposal of spent or used fuels - the uranium still decaying into harmful subatomic particles (radiations). Improper nuclear-waste storage and disposal result in environmental contamination. Further, there is a risk of accidental leakage of nuclear radiation. The high cost of installation of a nuclear power plant, high risk of environmental contamination and limited availability of uranium makes large-scale use of nuclear energy prohibitive.

**4.b Nuclear Fusion:** Currently all commercial nuclear reactors are based on nuclear fission. But there is another possibility of nuclear energy generation by a safer process called nuclear fusion. Fusion means joining lighter nuclei to make a heavier nucleus, most commonly hydrogen or hydrogen isotopes to create helium.

$$ {}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \xrightarrow[\text{fusion}]{\text{nuclear}} {}^{4}_{2}\mathrm{He} + \text{energy} $$

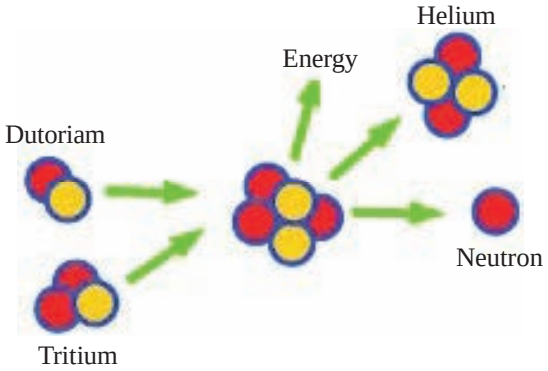



**Figure-8**

It releases a tremendous amount of energy, according to the Einstein equation, as the mass of the product is little less than the sum of the masses of the original individual nuclei.

Such nuclear fusion reactions are the source of energy in the Sun and other stars. It takes considerable energy to force the nuclei to fuse. The conditions needed for this process are extreme - millions of degrees of temperature and millions of Pascals of pressure.

**Benefits of nuclear energy:** 1. A small amount of U-235 can produce huge quantities of energy.
2. The reaction does not require oxygen.

Harmful effects of nuclear energy: The reaction produces radioactive substances which can cause great harm.

## 18.6 CREDA

CREDA stands for Chhattisgarh state Renewable Energy Development Agency and this is an enterprise under the department of energy of Chhattisgarh government,

Following are the objectives behind formation of CREDA:

1. To conserve energy - this means to ensure proper utilization of the energy sources and save energy.

2. To increase the use of non-conventional sources of energy in order to decrease the dependency on conventional sources.

3. Since the energy obtained from non-conventional sources of energy is environment friendly, therefore CREDA works to motivate people towards using these energy sources such as solar energy, wind energy and biomass energy.

The working models of conventional sources of energy for example, hydal power, wind power, solar energy power (solar car, solar hut, solar-boat) and biogas plants are displayed for demonstration purpose at Chhattisgarh energy education park (Raipur) managed by CREDA.

**In daily life, we should take care of the following points to ensure proper utilization of energy-**

1. People should be encouraged to use non-conventional sources of energy.
2. Pressure cookers should be used for cooking food.
3. Appliances which can be operated using solar energy like solar cooker, solar heaters, solar panels etc. should be used.
4. Biogas as a fuel should be promoted is rural areas.
5. To save valuable fuels like petrol, cycles should be used to travel short distances.



## What we have learnt

- The law of conservation of energy is followed when energy gets converted/ transformed from one form to another.
- Energy has many forms. (There are various forms of energy).
- The process of burning of fuel is called combustion.
- The temperature of solar cooker box increases to 100-140° C in 2-3 hours.
- Bio gas is a mixture of methane, carbon dioxide, hydrogen and hydrogen sulphide where the main component is methane.
- Nuclear energy can be obtained using two nuclear processes - (i) Nuclear fusion (ii) Nuclear fission.
- CREDA stands for Chhattisgarh state Renewable Energy Development Agency, an enterprise under the department of energy of Chhattisgarh government.
- Solar cells directly convert solar energy to electric energy.
- Wind energy is used for doing mechanical work or to generate electricity.
- Flowing water is a source of hydel-energy, and it is used in production of hydro-electricity.

- The microbial decomposition of the dung of cows and other similar domestic animals is used to produce biogas.
- Coal, petroleum and natural gases all are fossils fuels.

## Keywords

Hydel-power energy, tidal energy, wave energy, ocean thermal energy, geothermal energy, biogas, nuclear energy, nuclear fission, nuclear fusion, chain reaction, hydrogen bomb, fossil fuels, CREDA.

## Exercise:

1. Choose the correct option:

(i) Most of the energy sources used by us utilize solar energy. Which energy source out of following is not derived from solar energy:

(a) Geo thermal energy (b) Wind energy

(c) Nuclear energy (d) Bio mass

(ii) Biogas is a mixture of following gases-

(a) Nitrogen, Helium, Hydrogen (b) Oxygen, Nitrogen, Hydrogen

(c) Ethane, Oxygen, Hydrogen (d) Methane, Carbon dioxide, Hydrogen

(iii) Nuclear fission is a reaction in which-

(a) Two light nuclei combine

(b) Breaking of a heavy nucleus into light nuclei.

(c) Both of the above. (d) None of the above.

(iv) Fossil fuels are-

(a) Coal, petroleum, natural gas (b) Biogas, wave energy, nuclear energy.

(c) Tidal energy, hydel power energy.

(d) Wave energy, ocean thermal energy, geo thermal energy.

(v) Solar cell is a device which converts-

(a) Electric energy into kinetic energy (b) Solar energy into potential energy

(c) Solar energy into electric energy (d) Solar energy into nuclear energy

2. Fill in the blanks-
   (i) We cannot use solar heaters on a ..... day to heat water.
   (ii) Lighting a torch is conversion of chemical energy into ..... energy.
   (iii) The energy generated due to oceans is .....
   (iv) A solar cooker converts ..... energy into ..... energy.
3. Explain the mechanism of solar cooker with the help of a labeled diagram,.
4. Explain why-
   (i) The inside of a solar cooker box is painted black.
   (ii) The solar cooker box is covered with a glass-sheet.
   (iii) Plane mirror reflectors are used inside the solar cookers
5. Name the device which converts solar energy into electric energy.
6. Apart from cattle-dung, which other material can be put inside a biogas plant?
7. Explain any two differences between nuclear fusion and nuclear fission.
8. On what basis we can classify the sources of energy into renewable and non-renewable.
9. Write two sources of energy which you think are renewable. Give reasons for your choices.
10. Write down the benefits of use of solar energy.
11. What are limitations of wind energy?
12. What do you understand by conversion of energy?